

ASSIGNMENT-2

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BRANCH- MECHANICAL
SEC- 50
SUBJECT- MII

Who was Aryabhatta? Briefly explain his contributions to mathematics.

- * Aryabhatta (476 BCE) was one of the greatest Indian mathematicians and astronomer.

* He introduced place value system

* He calculated π as 3.1416

* He proposed and developed trigonometry

Q2) Describe the structure of the Aryabhatiya and explain its four chapters.

Ans:- Aryabhatiya is a famous mathematician and astronomer text written by Aryabhatta. It consists of 121 verses divided into 4 chapters, each dealing with specific topics.

The four chapters are as follows:-

(1) Gitikapada

* Written in Gitika meter and containing 13 verses

* Introduces important astronomical constants and planetary revolutions.

* One yuga = 4,320,000 years

(2) Granthapada

* This is the mathematical heart of Aryabhatiya

* It includes topics:

→ Arithmetic operation

→ Geometry

→ Area & volume calculation

(3) Kalakriyapada

* Focuses on → * Measurement of time

* Calendar calculation

* Planetary periods

(4) Golapada

* Earth as a sphere

* Planetary motion

* Eclipses

Q3) What is meant by the term Siddhanta? Who wrote the Pancasiddhantika? Name five Siddhantas.

Ans:- The word Siddhanta means "established conclusion" or "settled doctrine".

Varahamihira wrote the Pancasiddhantika which was written around 6th century CE.

The five Siddhantas are:-

(i) Vasistha Siddhanta

(iv) Paulisa Siddhanta

(ii) Paitamaha Siddhanta

(v) Saura Siddhanta

(iii) Romaka Siddhanta

Q4) What is the Kuttaka method developed by Aryabhata? Explain its purpose.
 Ans:- Aryabhata developed a method called Kuttaka to solve indeterminate linear equations.

$$ax + by = c, \text{ where } a, b \text{ \& } c \text{ are integers}$$

Purpose

→ To find soln of equation where exact division isn't possible

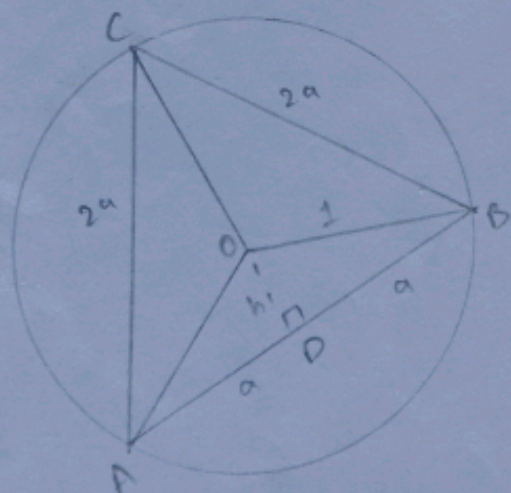
Q5) Explain Aryabhata's rule for π value with upto four decimal accuracy.

Ans:- The rule states that, Add 4 ~~to~~ 100, multiply by 8 and then add 62,000. The result is the circumference of a circle with diameter 20,000.

$$(100 \times 4) \times 8 + 62,000 = 62,832$$

$$\pi = \frac{\text{Circumference}}{\text{Diameter}} = \frac{62,832}{20,000} = 3.1416$$

Q6) Explain the Polygon doubling formula for computing π .



$$\begin{aligned} \text{Area of } \triangle OAB &= \frac{1}{2} \times h \times 2a \\ &= ha \end{aligned}$$

$$h = \frac{a}{\sqrt{3}}$$

$$h^2 + a^2 = 1^2$$

$$\frac{a^2}{3} + a^2 = 1$$

$$\Rightarrow \frac{4a^2}{3} = 1$$

$$\Rightarrow a^2 = \frac{3}{4} \Rightarrow a = \frac{\sqrt{3}}{2}$$

$$\text{Perimeter} = 3\sqrt{3}$$

$$\frac{P}{D} = \frac{3\sqrt{3}}{2} = 2.60$$

* Polygon doubling method is a classical method used to approximate the circumference of a circle and hence value of π is found.

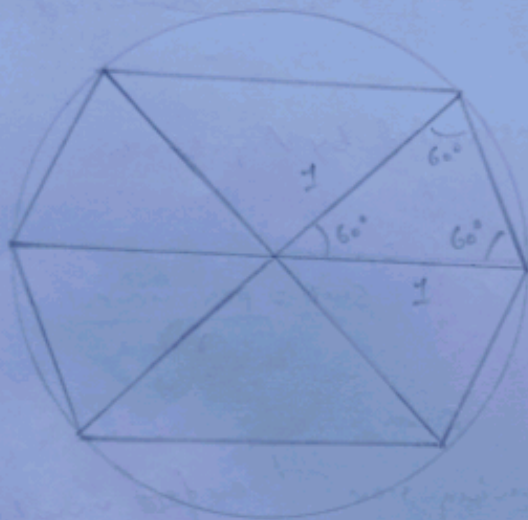


$$b^2 = 1 + 1 = 2$$

$$b = \sqrt{2}$$

$$\text{Perimeter (P)} = 4\sqrt{2}$$

$$N = 4 \rightarrow 2.83$$



$$180^\circ = 2\theta + 60^\circ$$

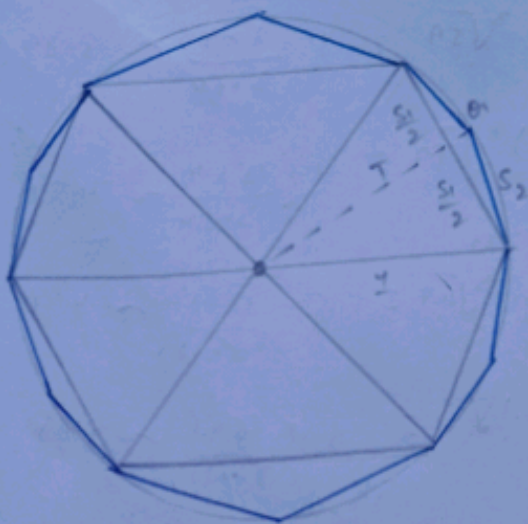
$$120^\circ = 2\theta$$

$$\text{or } \theta = 60^\circ$$

$$\text{Perimeter (P)} = 6$$

$$\frac{P}{D} = \frac{6}{2} = 3$$

$$N = 6 \rightarrow 3$$



$$S_1 = 1$$

$$T = \sqrt{1 - \left(\frac{S_1}{2}\right)^2}$$

$$P = NS_1$$

$$a = 1 - T$$

$$S_2 = \sqrt{a^2 + \left(\frac{S_1}{2}\right)^2}$$

$S_1 \rightarrow$ Side of 6-sided polygon

$S_2 \rightarrow$ Side of 12-sided polygon

Q7) Describe Aryabhata recursive sine difference formula.

Ans:- $\delta S_1 = \epsilon$

$$\delta S_m = \delta S_{m-1} - (1/\epsilon)$$

$$x \leq \delta S_i \quad (i=1 \text{ to } m-1)$$

* Linear recursion for non linear function

Q8) Discuss Aryabhata's contribution to trigonometry & sine tables.

Ans:- * The development trigonometry was done by the Aryabhata (476-550)

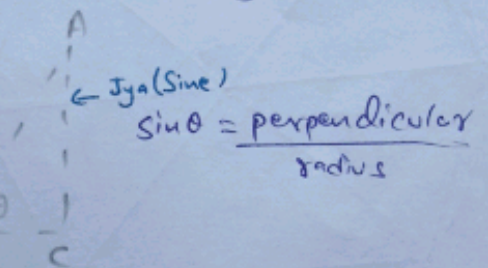
* His contribution transformed earlier geometry ideas into numerical & computational trigonometry.

* The first systematic sine table was constructed by Aryabhata in his Aryabhatiya (499 CE).

* Tabular for 0 to $\pi/2$

* Uses sine differences

* Step size $\pi/48$ (225 minutes)



Q9) Calculate the cube roots of the following non perfect number using Vedic mathematics.

Ans: (i) $\sqrt[3]{29} = 3 + \frac{2 \times 3}{3 \times 27} \sim 3.07$ (v) $\sqrt[3]{59} = 4 - \frac{5 \times 4}{3 \times 64} \sim 3.89$

(ii) $\sqrt[3]{53} = 4 - \frac{11 \times 4}{3 \times 64} \sim 3.77$

(vi) $\sqrt[3]{79} = 4 + \frac{15 \times 4}{3 \times 64} \sim 4.31$

(iii) $\sqrt[3]{38} = 3 + \frac{11 \times 3}{3 \times 27} \sim 3.407$

(vii) $\sqrt[3]{107} = 5 - \frac{18 \times 5}{3 \times 125} \sim 4.76$

(iv) $\sqrt[3]{67} = 4 + \frac{3 \times 4}{3 \times 64} \sim 4.06$

(viii) $\sqrt[3]{131} = 5 + \frac{16 \times 5}{3 \times 125} \sim 5.08$

Q10) Calculate the cube root of the following perfect number being used in Vedic mathematics.

(i) $\sqrt[3]{9261} \rightarrow 21$

(iii) $\sqrt[3]{6859} = 19$

(ii) $\sqrt[3]{74088} \rightarrow 42$

(iv) $\sqrt[3]{50653} = 37$

$$(v) \sqrt[3]{389017} = 73$$

$$(vi) \sqrt[3]{42875} = 35$$

$$(vii) \sqrt[3]{-94192} = -48$$

$$(viii) \sqrt[3]{-226981} = -61$$

$$(ix) \sqrt[3]{438976} = 76$$

$$(x) \sqrt[3]{-531441} = -81$$

Q11) Find out the square root of the following number using vedic mathematics

(i)

	4	8	.	7	5	
2	2	3	7	6	.	5920
	-1	6				
88		7	7	6		
		-7	0	4		
967			7	2	59	
			-6	7	69	
9745				4	9020	
				-4	8725	
					295	

(iv)

	0	.	9	4	5
0	0	.	8	9	31
	-0				
09		0	8	9	
		-8	1		
184			8	3	1
			-7	8	6
1885				9	500
				-9	425
					75

(ii)

	1	2	6
1	1	5	876
	-1		
22		0	58
		-4	4
126			1476
			-1476
			0

(v)

	1	.	8	9	2
1	3	.	5	80210	
	-1				
28		2	5	8	
		-2	2	4	
369			3	402	
			-3	321	
3782				8110	
				-7	564
					546

(iii)

	3	5	4	
3	1	2	5316	
	-9			
65		3	53	
		-3	25	
704			2816	
			-2	816
				0

(vi)

	1	9	.	2	9
1	3	72	.	4318	
	-1				
29		2	4	2	
		-2	6	1	
382			1	143	
			-7	64	
3849				37918	
				-3	4641
					3277

Q12) Using Kuttaka method Euclidean Algorithm find out a solution to the Diophantine equation.

$$47x + 30y = 1$$

$$47 = 30(1) + 17$$

$$30 = 17(1) + 13$$

$$17 = 13(1) + 4$$

$$13 = 4(3) + 1$$

Solve for remainder;

$$17 = 47(1) + 30(-1)$$

$$13 = 30(1) + 17(-1)$$

$$4 = 17(1) + 13(-1)$$

$$1 = 13(1) + 4(-3)$$

$$1 = 13(1) + 4(-3) \quad \text{Substitute by 4}$$

$$= 13(1) + [17(1) + 13(-1)](-3)$$

$$= 13(1) + 17(-3) + 13(3)$$

$$= 13(4) + 17(-3) \quad \text{Substitute by 13}$$

$$= [30(1) + 17(-1)](4) + 17(-3)$$

$$= 30(4) + 17(-4) + 17(-3)$$

$$= 30(4) + 17(-7) \quad \text{Substitute for 17}$$

$$= 30(4) + [47(1) + 30(-1)](-7)$$

$$= 30(4) + 47(-7) + 30(7)$$

$$= 30(11) + 47(-7)$$

$$x = -7, y = 11$$

Q14) Ekanyanena Purvena

$$(i) 9999 \times 2378 = 23777622$$

$$(9-2)(9-3)(9-7)(9-7)$$

$$(ii) 999 \times 324 = 323676$$

$$(9-3)(9-2)(9-3)$$

Q15) Anurupyena (proportionality)

$$(a) 46 \times 44$$

$$\begin{array}{r} 46 \times 44 \\ \hline 50 \quad 24 \end{array}$$

$$= 2024$$

Working base = 40

Multiplication base = 4×10

$$50 \times 4 = 200 + 2 = 202$$

$$(b) 48 \times 42$$

$$\begin{array}{r} 48 \quad + 8 \\ \times 42 \quad + 2 \\ \hline 50 \quad 16 \end{array}$$

$$= 2016$$

Working base
Multiplication base

$$50 \times 4 = 200$$

$$(c) 41 \times 43$$

$$\begin{array}{r} 41 \quad + 1 \\ \times 43 \quad + 3 \\ \hline 44 \quad 03 \end{array}$$

$$= 1763$$

Working base = 40

Multiplication base = 4×10

$$44 \times 4 = 176 + 0 = 176$$

1) Find the positive square root for each of the following.

i)

$$\begin{array}{r} 625 \\ 6 \overline{) 390625} \\ \underline{-36} \\ 122 \\ \underline{1245} \\ 1245 \\ \underline{1245} \\ 0 \\ \underline{0} \\ 000 \end{array}$$

(iv)

	4	1	1	1						
1	1	2	3	4	3	2	1			
	<u>-1</u>									
2	<u>1</u>									
	<u>-</u>	0	2	3						
		<u>-</u>	2	1						
			<u>-</u>	2	4	3				
				<u>-</u>	2	2	1			
					<u>-</u>	2	2	2	1	
						<u>-</u>	2	2	2	1
							<u>-</u>	0		

ii)

$$\begin{array}{r} 534 \\ 5 \overline{) 285156} \\ \underline{25} \\ 103 \\ \underline{1064} \\ 1064 \\ \underline{1064} \\ 0 \\ 368 \end{array}$$

$$\begin{array}{r} 177 \\ 31329 \\ -1 \\ \hline 213 \\ -189 \\ \hline 2429 \\ -2429 \\ \hline 0 \end{array}$$

$$\begin{array}{r} 3 \\ 66 \\ 728 \\ \hline \end{array} \quad \begin{array}{r} 135424 \\ - 9 \\ \hline 454 \\ - 396 \\ \hline 5824 \\ - 5824 \\ \hline 0 \end{array}$$

(18) Write the name of the following number system mentioned in the vedic literature, 10^7 , 10^2 , 10^{11} , 10^{12} , 10^{14} , 10^{16} , 10^{17} , & 10^{53}

 $10^7 \rightarrow \text{Koti}$
$$10^9 \rightarrow \text{Asbuta}$$

10" → Kharva

$10^{12} \rightarrow$ Nisharua

$10^{14} \rightarrow$ Mahapadma

$10^{16} \rightarrow$ Shaukha

$10^{17} \rightarrow$ Talandhi

$10^3 \rightarrow$ Asankhyeya

Q13) For each of following linear Diophantine equation, either find the form of a general equation or show there are no integer solution.

(a) $21x + 14y = 147$

$\gcd(21, 14) = 7$

Since 7 divides 147
integer soln exist

(b) $30x + 47y = -11$

$47 = 30(1) + 17$

$30 = 17(1) + 13$

$17 = 13(1) + 4$

$13 = 4(3) + 1$

$\gcd(30, 47) = 1$

Since 1 divides -11
it has the integer solution

(c) $21x + 14y = 146$

$\gcd(21, 14) = 7$

Since 7 does not divide
146, there are no
integer soln

(d) $30x + 47y = 2$

$\gcd(30, 47) = 1$

Since 1 divides 2
it has the general solution

(e) $4x - 6y = 77$

$\gcd(4, 6) = 2$

Since 2 does not divide 77,
there are no integer soln.

(f) $4x - 6y = 78$

$\gcd(4, 6) = 2$

Since 2 divide 78,
there it has the integer soln.

16. Compute the square of the following integers:-

(i) 149

	5 th	4 th	3 rd	2 nd	1 st
digit=1	$1^2 = 1$	$2 \times 1 \times 4 = 8$	$2 \times 1 \times 9 = 18$		
digit=4	$\Rightarrow$	$\Rightarrow$	$4^2 = 16$	$2 \times 4 \times 9 = 72$	
digit=9			$\Rightarrow$	$\Rightarrow$	$9^2 = 81$

digit ^{1st}	digit ^{2nd}	digit ^{3rd}	digit ^{4th}	digit ^{5th}
			8	1
		8	1	
		6	1	
	2	7		
1	8			
2	0	2	2	2

(ii) 642

	5 th	4 th	3 rd	2 nd	1 st
digit=6	$6^2=36$	$2 \times 6 \times 4=48$	$2 \times 6 \times 2=24$		
digit=4	$\Rightarrow$	$\Rightarrow$	$4^2=16$	$2 \times 4 \times 2=16$	
digit=2			$\Rightarrow$	$\Rightarrow$	$2^2=4$

6 th	5 th	4 th	3 rd	2 nd	1 st
3	6				
	4	8			
		2	4		
		1	6		
			1	6	
					4
4	1	2	1	6	4

(iii) 1777

	7 th	6 th	5 th	4 th	3 rd	2 nd	1 st
digit = 1	$1^2 = 1$	$2 \times 1 \times 7 = 14$	$2 \times 1 \times 7 = 14$	$2 \times 1 \times 7 = 14$			
digit = 7	$\Rightarrow$	$\Rightarrow$	$7^2 = 49$	$2 \times 7 \times 7 = 98$	$2 \times 7 \times 7 = 98$		
digit = 7			$\Rightarrow$	$\Rightarrow$	$7^2 = 49$	$2 \times 7 \times 7 = 98$	
digit = 7					$\Rightarrow$	$\Rightarrow$	$7^2 = 49$

7 th	6 th	5 th	4 th	3 rd	2 nd	1 st
1						
1	4					
	1	4				
		1	4			
	4	9				
		9	8			
			9	8		
			4	9		
				7	8	
					4	9
3	1	5	7	7	2	9